Activity 2.8 – Memory and Watchdog Timer

1.Design Approach:

Designed RAM, ROM and watchdog timer modules using Verilog HDL. RAM supports read and write operations, ROM provides stored data, and the watchdog timer generates a timeout signal when the programmed count is reached.

2.Verification Steps:

- Created a separate testbench.
- Tested RAM write and read operations.
- Verified ROM data retrieval.
- Applied reset to the watchdog timer.
- Tested watchdog counter operation.
- Checked timeout signal generation.
- Simulated using ModelSim.

3.Results:

The testbench checks memory read/write operations and watchdog timeout generation. PASS/FAIL messages are displayed.

4.Issues Faced:

Memory initialization must be handled correctly. Read and write operations require proper synchronization. Watchdog counter limits must be configured accurately.

5.Conclusion:

Successfully designed and verified RAM, ROM and watchdog timer modules using Verilog HDL.